

# Mikołaj Mieszko Charchuta

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## PROFILE

**Bioinformatician turning exploratory genomics into reproducible, production-grade pipelines.** I build containerised, re-runnable analyses (Nextflow, Snakemake, Docker, Conda) and specialise in chromosome conformation capture (Meta Hi-C, 3C-seq) and machine learning on bacterial pangenomes. At Phagos I turn Meta Hi-C data into phage–host pairings for therapeutic discovery. At Adam Mickiewicz University (AMU) I run an independently grant-funded project on the 3D genome of *Bacillus thuringiensis* and mine its pangenome for Cry protein toxins. Earlier I worked in personalised medicine, cellular agriculture and commercial biosafety analytics, so I am used to QA standards, turnaround times and teams that mix wet lab and code. Three publications and invited teaching on reproducibility; EU remote-first, open to B2B contracts and industrial PhD (CIFRE) partnerships.

## EXPERIENCE

### **PHAGOS** | Bioinformatician
Jul 2026 -- Present | Remote / Paris, FR

Phage-therapy company

- Deliver phage–host association calls into the therapeutic discovery pipeline by analysing Meta Hi-C data end-to-end, from raw reads to ranked pairings.
- Made analyses re-runnable and free of environment drift by rebuilding ad-hoc scripts as containerised Nextflow workflows (Docker, Conda) used across the R&D team.
- Guide two research projects and joint analyses with external academic partners (incl. Institut Pasteur), aligning experimental design with computational feasibility.

### **AMU, DEPT. OF MICROBIOLOGY** | Contract Researcher
Sep 2025 -- Jan 2028 | Remote / Poznań, PL

Genomes of insect pests and biopesticidal microbes

- Won competitive Study@Research funding (app. nr. 201/34/UAM/0074) to run an independent project on the 3D genome of *B. thuringiensis*, owning Hi-C library prep, sequencing and analysis solo.
- Produced the dataset behind a first-author manuscript by mining the *B. thuringiensis* pangenome for Cry9 toxin determinants, linking gene variants to predicted protein features.
- Combine comparative genomics with predictive protein design for insecticidal Cry toxins.
- Standardised the group's Hi-C/3C-seq analysis with reusable containerised workflows for contact-map QC and scaffolding.

### **GESELLSCHAFT FÜR BIOANALYTIK** | Scientist
Jul 2024 -- Jun 2026 | Poznań, PL

Commercial bioanalytics laboratory

- Delivered allergen and biosafety analyses to commercial turnaround and QA standards.
- Covered the whole chain from field sampling through lab work to computational analysis and client reporting.

### **BENE MEAT TECHNOLOGIES** | Bioinformatician
Mar 2024 -- Jul 2024 | Prague, CZ

Cultivated pet food manufacturer

- Provided end-to-end bioinformatics support to cultivated-meat cell-line R&D.
- Ran transcriptomics (DESeq2, rMATS, rnaSPAdes) and nanopore analyses.
- Automated bioreactor workflows, connecting bioprocess runs to downstream analysis.

### **LEIDEN UNIVERSITY MEDICAL CENTER** | Bioinformatician
Jun 2023 -- Oct 2023 | Leiden, NL

Endocrinology Department

- Contributed to two peer-reviewed publications by building the pipelines and figures for a murine MASLD study.
- Provided bioinformatics support and data visualisation to the lab.
- Conducted animal research, including work with murine tissues and organs.

### **MNM DIAGNOSTICS** | Genome Analyst
Feb 2022 -- Jul 2022 | Poznań, PL

Personalised-oncology start-up

- Interpreted clinical whole-genome data for personalised oncology, reporting variants to clinicians.
- Supported the machine-learning team with feature extraction from methylation (WGBS) data, the data lake and explainable-AI (XAI) work.
- Automated tasks and data acquisition, and worked with biobanks under strict patient-data safety rules.

### **DANUBE INTERNATIONAL SCHOOL VIENNA** | Lab Assistant
Jun 2026 -- Aug 2028 | Vienna, AT

- Design and run hands-on science labs; mentor students through independent research projects.

## RESEARCH PROJECTS & GRANTS

**3D GENOME OF B. THURINGIENSIS** | Principal investigator — 2026 -- 2027 | Study@Research, AMU

- Independent, competitively funded project; Hi-C library preparation, sequencing and analysis carried out solo.

**DYNAMAP NANOPORE PROFILING** | Team lead — Oct 2023 -- Oct 2025 | AMU grant for talented students

- Led a student team in funded BSc thesis projects benchmarking a new nanopore-based microbiome profiling technique against established standards, showing its potential to lower costs.

**SPUN GLOBAL SOIL MAPPING** | Researcher — Nov 2022 -- Apr 2024 | International grant

- Predicted soil biodiversity hotspots with machine learning on Illumina data for a global mapping project (field video).

**TREES OF THE FUTURE** | Team member — Oct 2021 -- Oct 2023 | AMU team grant

- Identified interaction hubs between soil microorganisms using Ion Torrent sequencing.
- Awarded best poster at the national metagenomics symposium; article in preparation.

**NAÏVER** | Open-source R package — github.com/mmcharchuta/NaiveR

- Naïve methods for taxonomic classification from marker genes, released from my BSc thesis.

## EDUCATION

**BOKU UNIVERSITY, VIENNA** | MSc Bioinformatics (Erasmus exchange) — Sep 2026 -- Jul 2027

Thesis project with Prof. Christian Stauffer.

**ADAM MICKIEWICZ UNIVERSITY, POZNAŃ** | MSc in Bioinformatics — Oct 2025 -- Jul 2027

Thesis: *In silico analysis of Cry9 toxins and their genetic determinants in the Bacillus thuringiensis pangenome* (first-author article in preparation).

**ADAM MICKIEWICZ UNIVERSITY, POZNAŃ** | BSc in Bioinformatics & Biotechnology — Oct 2021 -- Sep 2025

Thesis: *naïve methods for taxonomic classification from marker genes*, released as the NaïveR R package.

**CHARLES UNIVERSITY, PRAGUE** | Erasmus+ research exchange — Winter 2022/23 • Summer 2023/24

- Institute of Microbiology, Czech Academy of Sciences: stable isotope probing of polar soil microbes for carbon tracking; meta-omics.
- BIOCEV: alternative splicing analysis (rMATS) on human RNA-seq data.
- CryoEM of cultivated-meat proteins.

**UNIVERSITY OF GDAŃSK & MEDICAL UNIVERSITY OF GDAŃSK** | National exchange — Winter 2023/24

- Metabolomics and proteomics of bioprotective synthetic microbial communities (SynComs).
- Phage genomics.

## PUBLICATIONS

**In silico analysis of Cry9 toxins and their genetic determinants in the *Bacillus thuringiensis* pangenome. Charchuta, M. M.** (first author), Baranek, J., et al. *In preparation, 2026.*

**Combination of dietary fiber and exercise training improves fat loss in mice but does not ameliorate MASLD more than exercise alone.** Kovynev, A.; **Charchuta, M. M.**; et al. *Am. J. Physiol. Gastrointest. Liver Physiol.*, 2025.

**WED-517 Oatmeal a day does not keep the doctor away: inulin and exercise impact on MASLD amelioration and gut-liver crosstalk.** Kovynev, A.; **Charchuta, M.**; et al. *J. Hepatol.*, 2024.

## TALKS, TEACHING & COMMUNITY

**CRABS 2026 / HEALTH INTELLIGENCE** | Invited Speaker & Workshop Lead — Aug 2026 | Kyiv, UA

- Lecture on *Reproducibility in Computational Genomics*.
- Two hands-on Nextflow pipeline workshops.
- Talk on Meta Hi-C and machine learning for bacterial genomes.

**BIOINFORMATICS COMMUNITY PL** | Co-organiser — 2026 -- Present

- Event coordination, website and open-science advocacy.

## SKILLS

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|---|---|
| **Reproducibility** | Nextflow, Snakemake, Docker, Conda, Git, CI/CD, Slurm, AWS |
| **Programming & ML** | R, Python, Bash, C++; scikit-learn, tidymodels; R package development |
| **Genomics** | 3D genomics (Hi-C, Meta Hi-C, 3C-seq), pangenomics, WGS, WGBS, ChIP-seq; samtools, GATK, IGV |
| **Phages & proteins** | Phage–host association (Meta Hi-C), toxin and gene mining in pangenomes, predictive protein design, structure analysis (PyMOL), proteomics (PEAKS Studio, ProteinPilot) |
| **Transcriptomics** | DESeq2, rMATS, rnaSPAdes |
| **Metagenomics** | QIIME2, mothur, NaïveR, stable isotope probing |
| **Sequencing & wet lab** | Nanopore, Illumina, Ion Torrent; Hi-C library preparation, qPCR, DNA isolation, animal work |

## LANGUAGES

English (C2) • Polish (native) • German (B2) • French (B2) • Czech (A2) • Hungarian (A2)

## INTERESTS

Beekeeping and fungiculture • bikepacking • vegan cooking

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